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# Berry Phases and Curvatures in Electronic-Structure Theory

David Vanderbilt  
Rutgers University

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### Rahman prize for:

- Theory of polarization (King-Smith & Vanderbilt)
- Ultrasoft pseudopotentials

### Three quick preliminaries:

- Who was Aneesur Rahman?
- Who is Dominic King-Smith?
- A parable about referee reports...

# Who was Aneesur Rahman?

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*"Father of Molecular Dynamics"*

- Born Hyderabad, India
- Educ. Cambridge, Louvain
- Argonne Natl. Labs 1960-85
- U. Minnesota 1985-87
- Died 1987
- Rahman Prize established in 1992 with funds from IBM

Photo courtesy Sam Bader via Marie-Louise Saboungi

# Who is Dominic King-Smith?

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## *"Father of Bettina"*

- PhD, Cambridge, UK
- Postdoc at Rutgers `91-`93
- Biosym/MSI/Accelrys `93-`01
- Presently at:



Accelrys Job title:  
"Product Manager, Quantum Mechanics"



# Ultrasoft Pseudopotentials

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Referee B

Referee's Report: Manuscript #LJ4237

Title: Soft self-consistent pseudopotentials in ...

Author: D. Vanderbilt

This paper is not suitable for publication in Phys. Rev. Letters. The basic idea the author presents is 30 years old, it is not clear that his reformulation is of any use, and technical refinements of pseudopotential theory are by now primarily of interest to specialists. A somewhat revised version should be suitable for publication as a brief report or regular paper in Phys. Rev. B. I will expand on these summary comments below.

The original Phillips-Kleinman pseudopotential was separable, linearly energy-dependent, and required the addition of all-electron terms to produce the correct charge density (J. C. Phillips and L. Kleinman, Phys. Rev. 118, 287 (1959), Eq. 2.8 in particular). While the construction of the author's "new" pseudopotential is much more complex, it shares these qualities as its principal distinguishing features from modern ab-initio pseudopotentials. The PK form with energy dependence is also exact, a quality which the author claims as a limit for his version. While the formal connection between the present work and PK is obscure, the qualitative features clearly indicate that this is a big step backwards. To be acceptable for publication in Phys. Rev. B, a revised version of this paper should refer to PK, acknowledge the similarities, and develop the formal connection to the extent possible.

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# Berry Phases and Curvatures in Electronic-Structure Theory

David Vanderbilt  
Rutgers University

# Introduction

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- By mid-1990s, density-functional perturbation theory allowed calculation of linear response to E-field
- However, it was not known how to:
  - Calculate **polarization itself**
  - Treat **finite E-fields**
- Analogous problem of calculating **orbital magnetization** also unsolved

# Introduction

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- Solutions of these problems are now in hand
  - Modern theory of polarization (1993)
  - Treatment of finite E-fields (2002)
  - Orbital magnetization (2005)
- Solutions rely heavily on two crucial ingredients:
  - Wannier functions
  - Berry phases and related quantities

*This talk:*

***Brief survey of methods!***

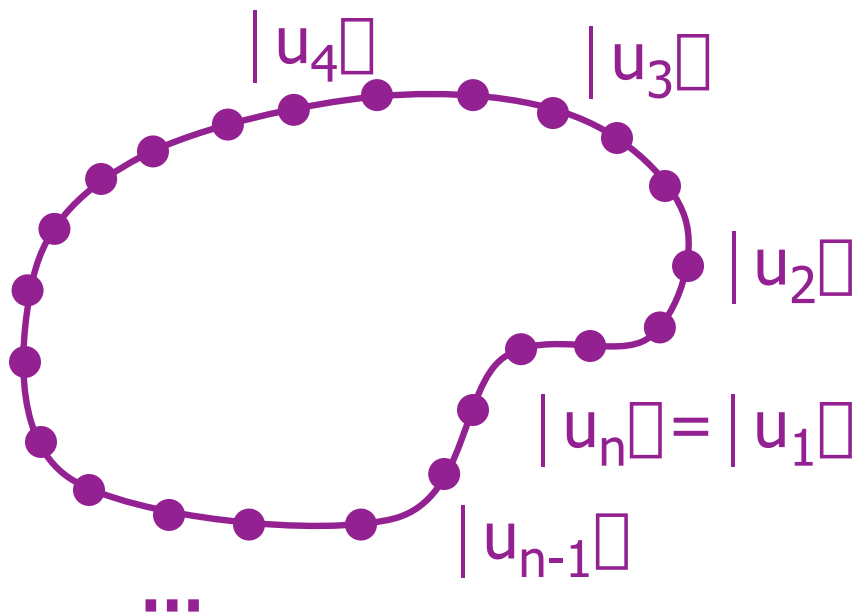
*Almost nothing on applications*

# Outline of Talk

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- Introduction
- Berry phases, potentials, and curvatures
- Realizations:
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  - Wannier functions
  - Electric fields
  - Anomalous Hall conductivity
  - Orbital magnetization
- Summary and prospects

# Berry phases

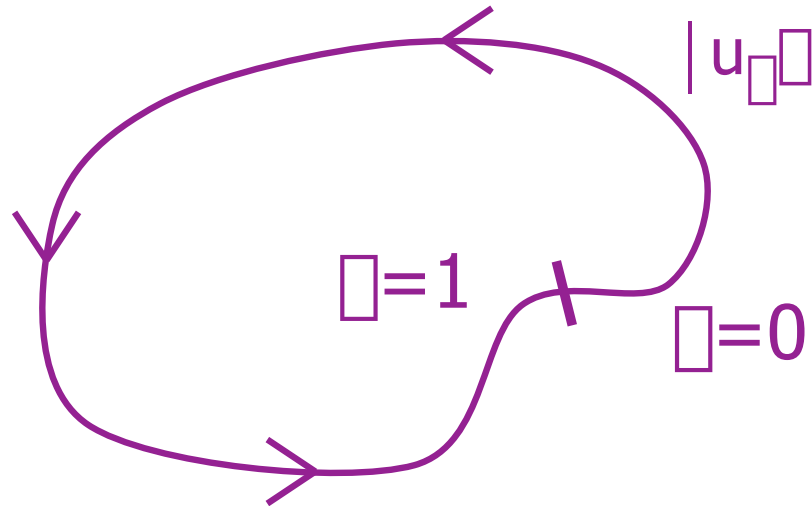


Now take limit  
that density of  
points  $\rightarrow \infty$

$$\phi = -\text{Im} \ln \left[ \langle u_1 | u_2 \rangle \langle u_2 | u_3 \rangle \dots \langle u_{n-1} | u_n \rangle \right]$$

Check:  $|\tilde{u}_2\rangle = e^{i\beta} |u_2\rangle$  has no effect.

# Berry phases



$$\phi = -\text{Im} \oint d\lambda \langle u_\lambda | \frac{du_\lambda}{d\lambda} \rangle$$

$$\phi = -\text{Im} \oint d\lambda \langle u_\lambda | \frac{d}{d\lambda} | u_\lambda \rangle$$

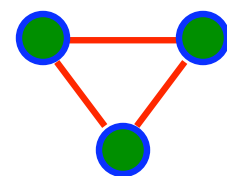
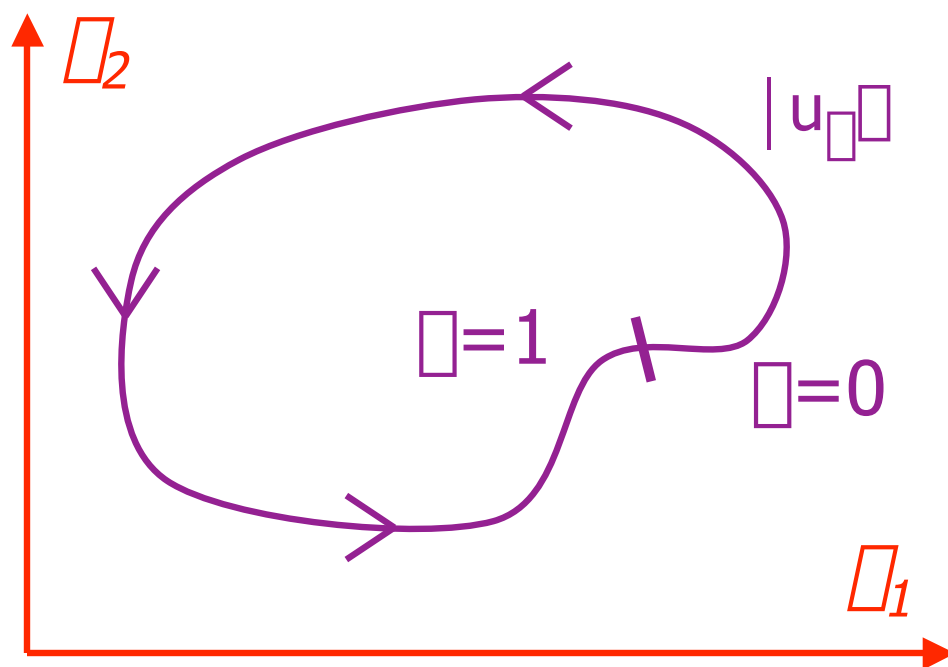
Continuum  
limit

Check:  $|\tilde{u}_\lambda\rangle = e^{i\beta(\lambda)} |u_\lambda\rangle$  has no effect.

[Follows using  $\beta(\lambda = 1) = \beta(\lambda = 0)$ .]



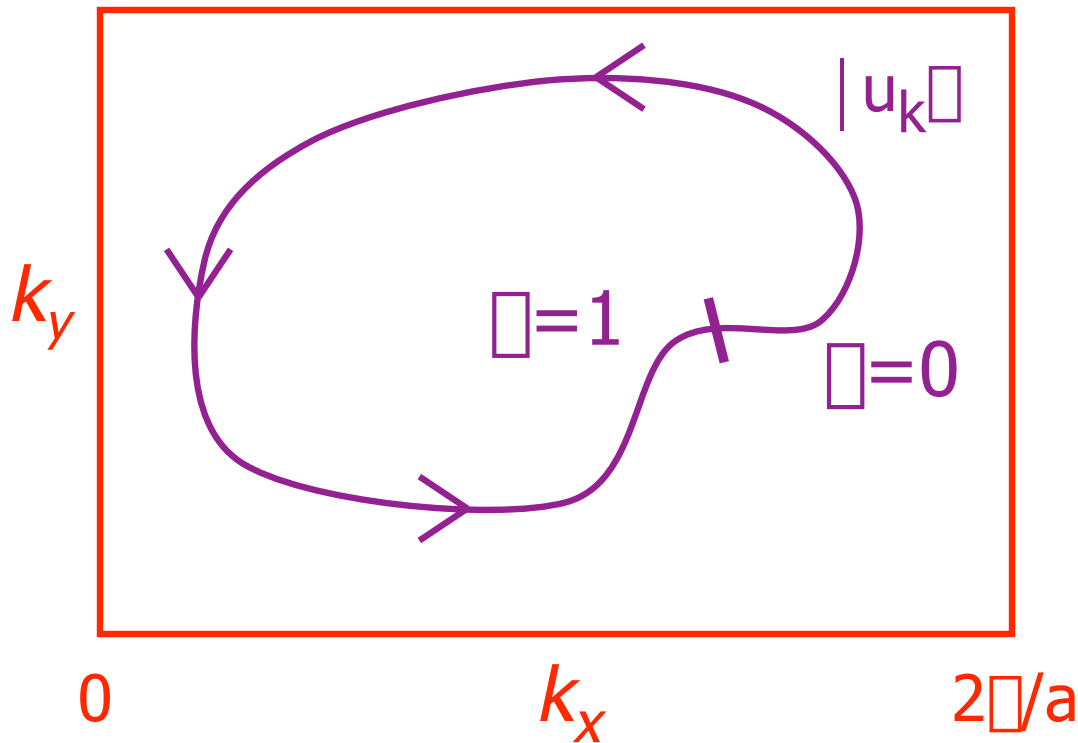
# (Context: Molecular coordinates)



$\text{Na}_3$

$(q_1, q_2)$

# Context: k-space in Brillouin zone

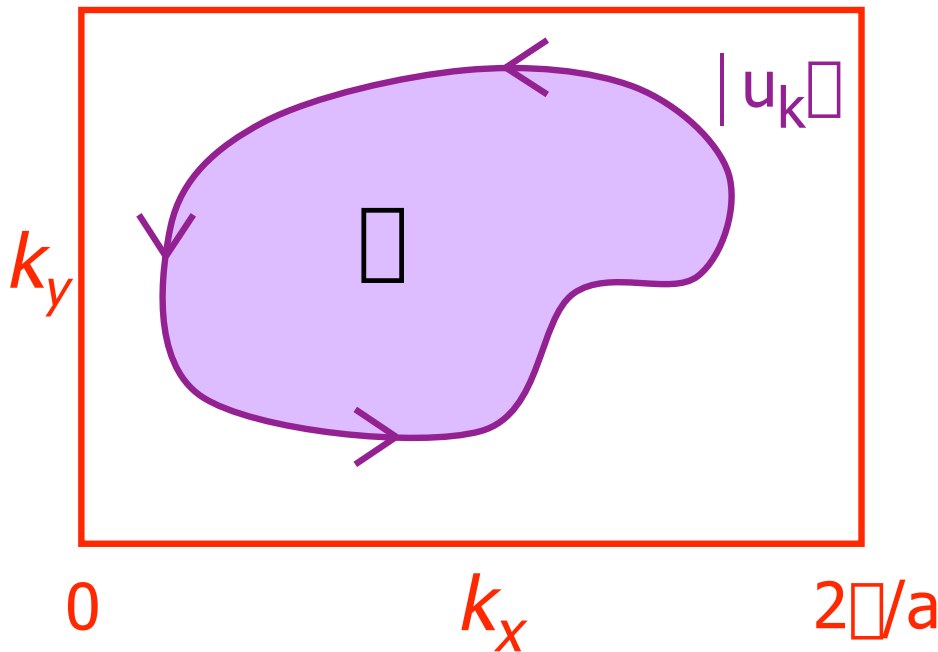


$$u_{\mathbf{k}}(\mathbf{r}) = e^{-i\mathbf{k}\cdot\mathbf{r}} \underbrace{\psi_{\mathbf{k}}(\mathbf{r})}_{\text{Bloch function}}$$

Bloch function

$$\phi = -\text{Im} \oint d\mathbf{k} \cdot \langle u_{\mathbf{k}} | \nabla_{\mathbf{k}} | u_{\mathbf{k}} \rangle$$

# Stokes theorem: Berry curvature



Berry potential:

$$\mathbf{A}(\mathbf{k}) = -\text{Im} \langle u_{\mathbf{k}} | \nabla_{\mathbf{k}} | u_{\mathbf{k}} \rangle$$

Berry phase:

$$\phi = \oint \mathbf{A}(\mathbf{k}) \cdot d\mathbf{k}$$

Berry curvature:

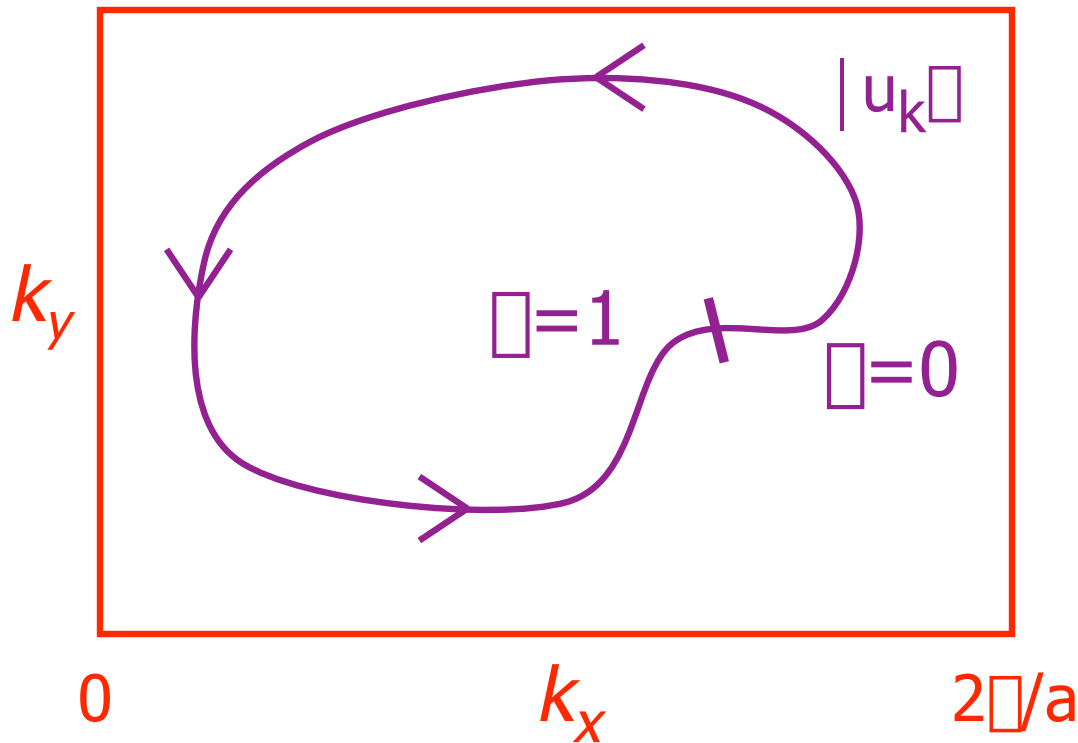
$$\Omega(\mathbf{k}) = \nabla \times \mathbf{A}$$

Stoke's theorem:

$$\phi = \int \Omega_z(\mathbf{k}) d^2k$$

$$\Omega_z(\mathbf{k}) = -2\text{Im} \left\langle \frac{du}{dk_x} \left| \frac{du}{dk_y} \right. \right\rangle$$

# Context: k-space in Brillouin zone

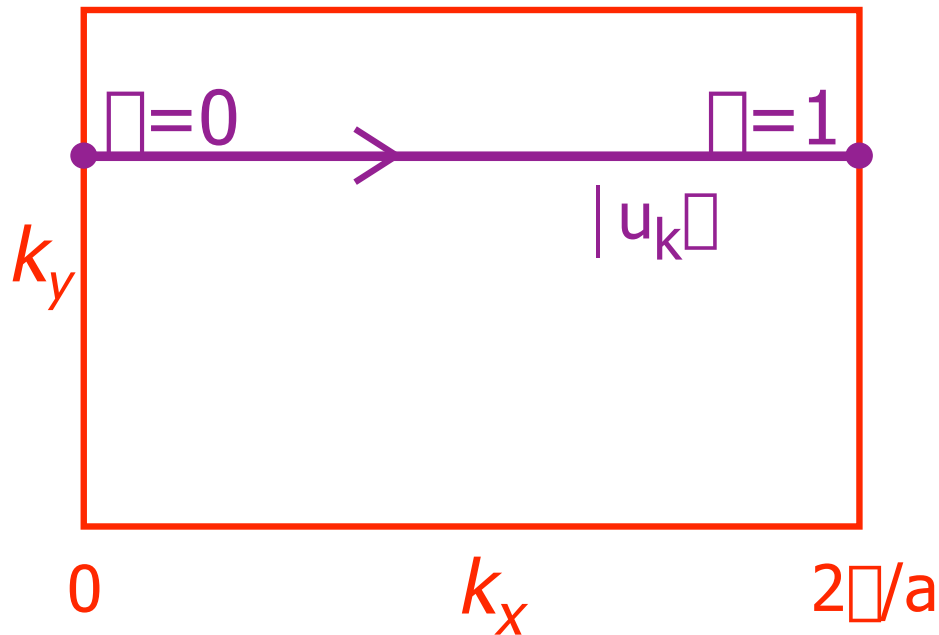


$$u_{\mathbf{k}}(\mathbf{r}) = e^{-i\mathbf{k}\cdot\mathbf{r}} \underbrace{\psi_{\mathbf{k}}(\mathbf{r})}_{\text{Bloch function}}$$

Bloch function

$$\phi = -\text{Im} \oint d\mathbf{k} \cdot \langle u_{\mathbf{k}} | \nabla_{\mathbf{k}} | u_{\mathbf{k}} \rangle$$

# Spanning the BZ



$$u_{\mathbf{k}}(\mathbf{r}) = e^{-i\mathbf{k} \cdot \mathbf{r}} \underbrace{\psi_{\mathbf{k}}(\mathbf{r})}_{\text{Bloch function}}$$

Bloch function

$$\phi = -\text{Im} \int dk_x \langle u_{\mathbf{k}} | \frac{d}{dk_x} | u_{\mathbf{k}} \rangle$$

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Does any of this  
have any connection  
to real physics  
of materials?

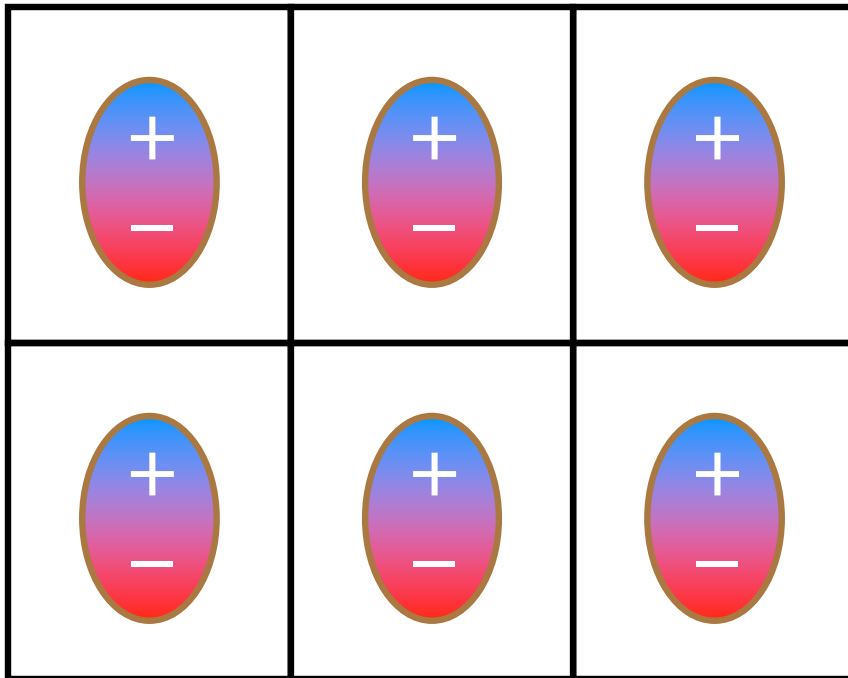
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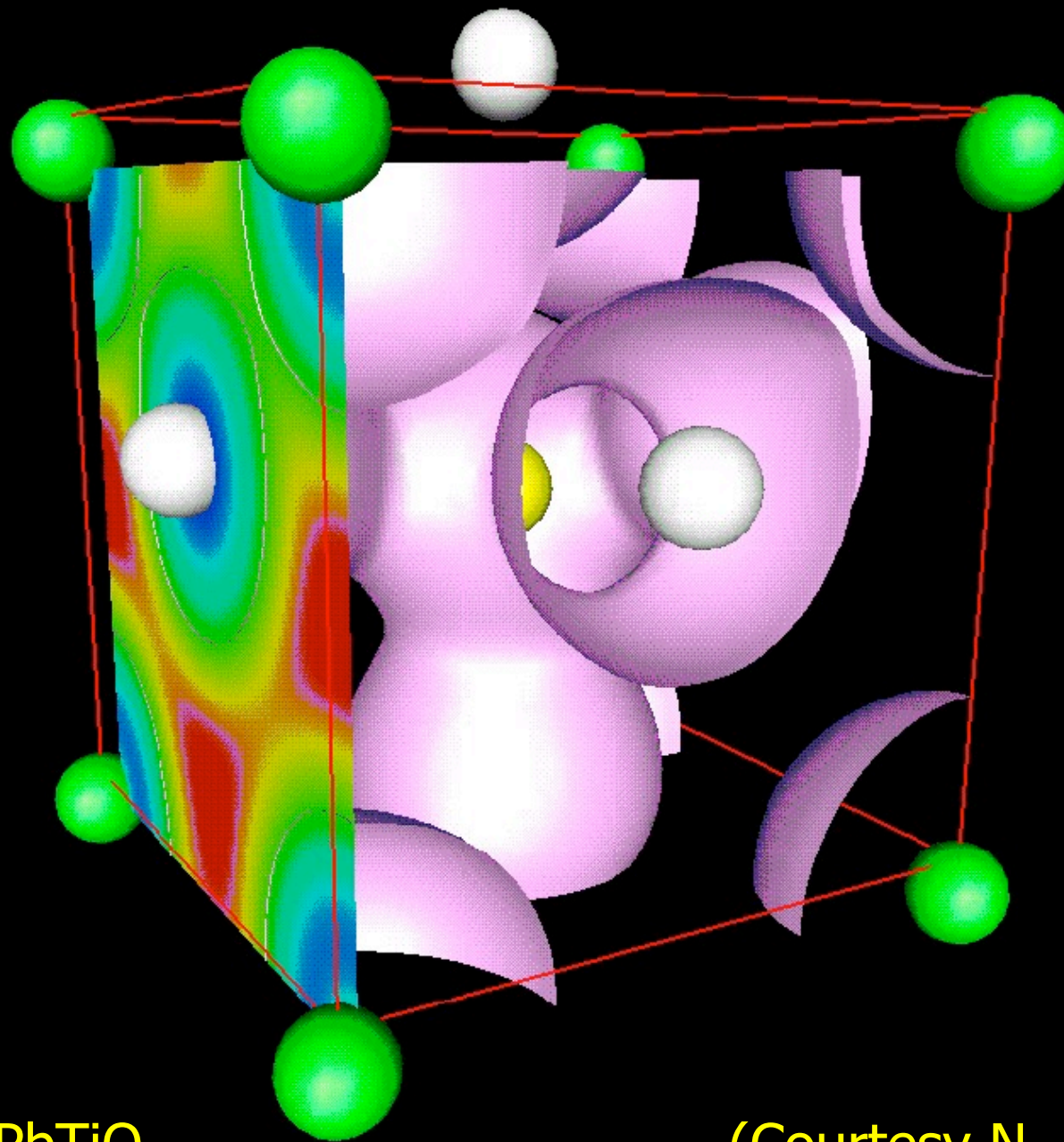
$$P = d_{cell} / V_{cell} ?$$

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- Textbook picture (Claussius-Mossotti)
- But does not correspond to reality!



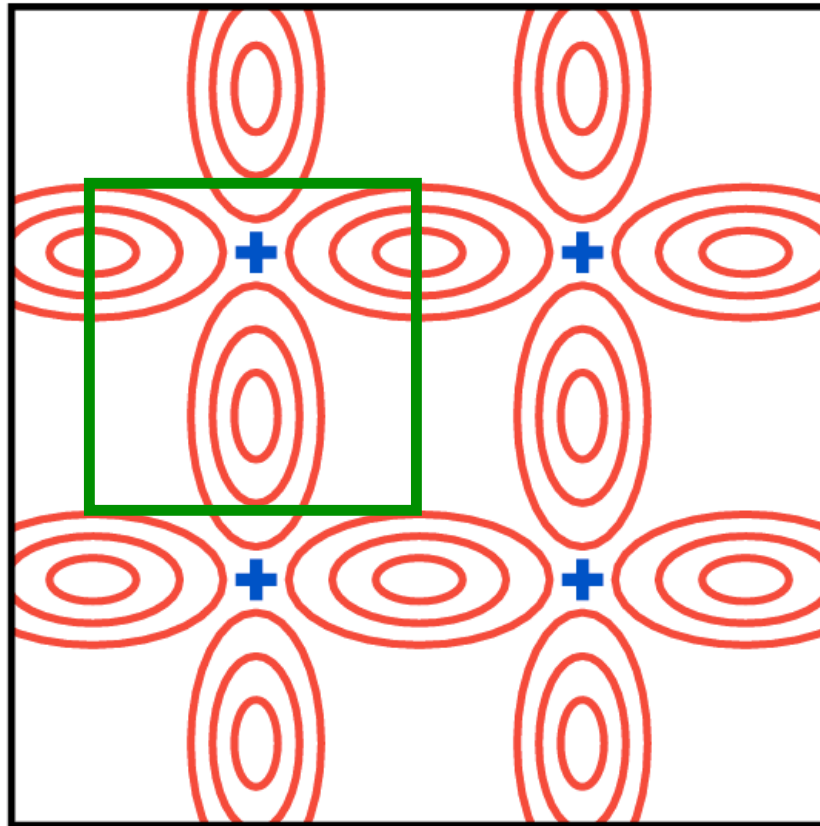


Ferroelectric  $\text{PbTiO}_3$

(Courtesy N. Marzari)

$$P = d_{cell} / V_{cell} ?$$

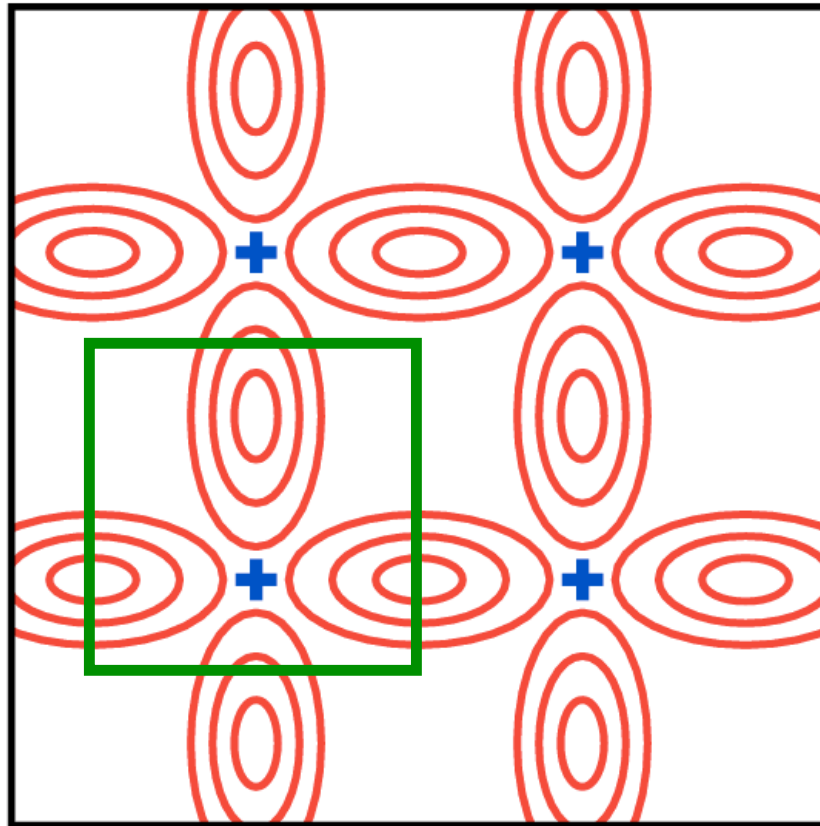
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$$d_{cell} = \uparrow$$

$$P = d_{cell} / V_{cell} ?$$

---



$$d_{cell} = \downarrow$$

# Berry-phase theory of electric polarization

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Use the **current**

$$\frac{d\mathbf{P}}{dt} = \frac{1}{V} \int_V d^3r \mathbf{j}(\mathbf{r})$$

instead of the **density**

$$\mathbf{P} = \frac{1}{V} \int_V d^3r \mathbf{r} \rho(\mathbf{r})$$

since  **$\mathbf{j}(\mathbf{r})$**  is periodic but  **$\mathbf{r} \rho(\mathbf{r})$**  is not.

Resta, 1992:

$$\Delta\mathbf{P} = \int \left( \frac{d\mathbf{P}}{dt} \right) dt$$

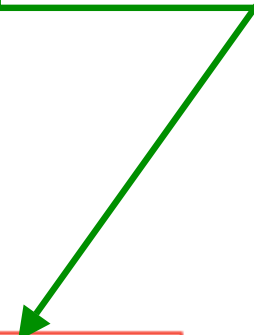
# Berry-phase theory of electric polarization

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Resta, 1992:

$$\Delta \mathbf{P} = \int \left( \frac{d\mathbf{P}}{dt} \right) dt$$

Berry potential!



King-Smith and Vanderbilt, 1993:

$$\Delta \mathbf{P} = \mathbf{P}(t_2) - \mathbf{P}(t_1) \quad \text{where}$$

$$\mathbf{P} = \frac{ie}{(2\pi)^3} \sum_n \int_{\text{BZ}} d^3k \langle u_{nk} | \nabla_{\mathbf{k}} | u_{nk} \rangle$$

$$\text{where } \psi_{nk}(\mathbf{r}) = e^{i\mathbf{k} \cdot \mathbf{r}} u_{nk}(\mathbf{r})$$

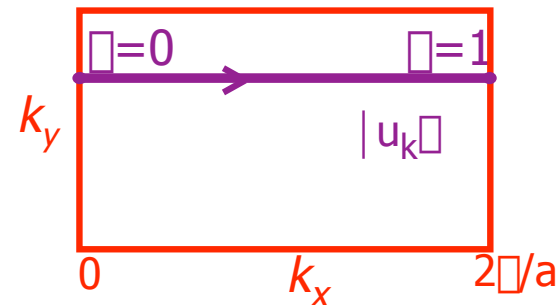
# Simplify: 1 band, 1D

$$\mathbf{P} = \frac{-e}{2\pi} \int_{\text{BZ}} dk \langle u_{\mathbf{k}} | i \frac{d}{dk} | u_{\mathbf{k}} \rangle$$

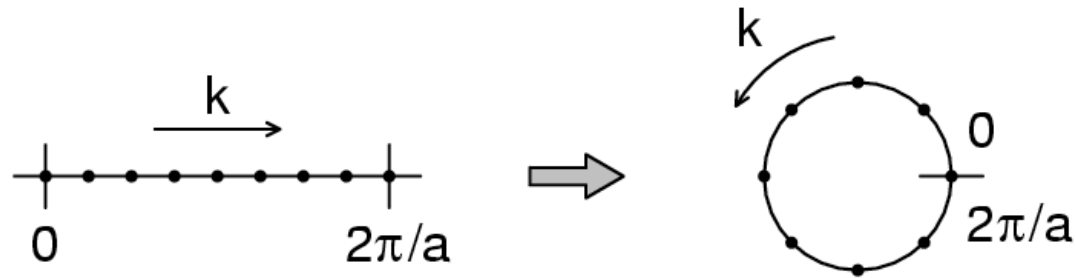
Heuristically,  $x \Leftrightarrow i \frac{d}{dk}$

$$\mathbf{P} = -e \frac{\phi}{2\pi} \quad \text{where} \quad \phi = \int_{\text{BZ}} dk \langle u_{\mathbf{k}} | i \frac{d}{dk} | u_{\mathbf{k}} \rangle$$

Berry phase!

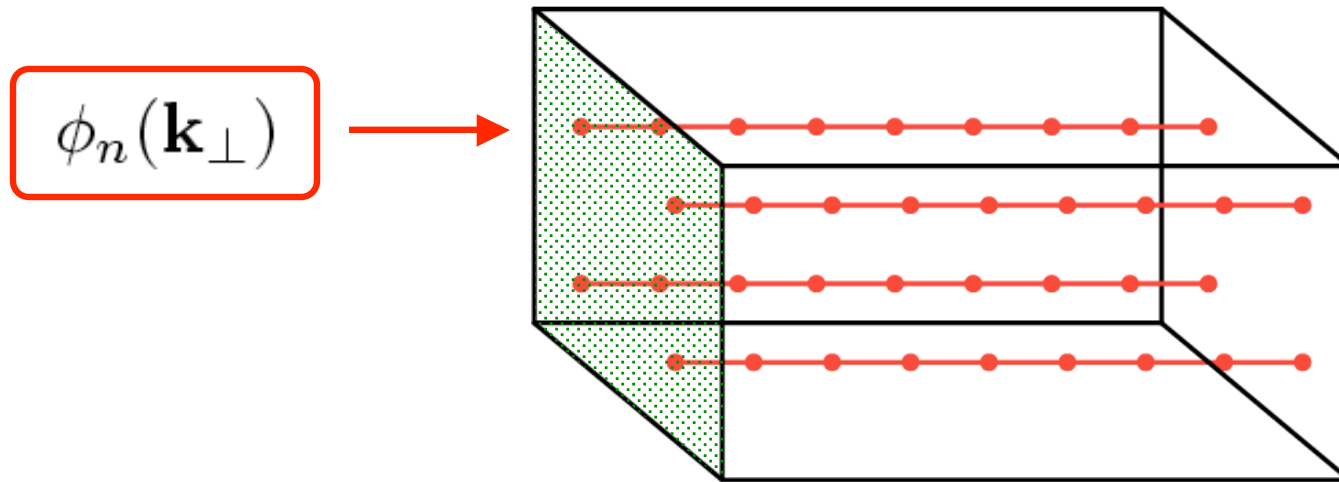


# Discrete sampling of k-space



$$\phi = \text{Im} \ln \prod \langle u_k | u_{k+\Delta k} \rangle$$

# Discretized formula in 3D

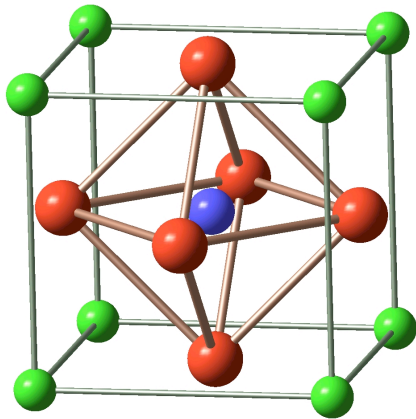


$$\mathbf{P}_n = \frac{1}{2\pi} \frac{e}{\Omega} \sum_j \phi_{n,j} \mathbf{R}_j \quad \text{where} \quad \phi_{n,j} = \frac{1}{N_{\mathbf{k}_\perp}} \sum_{\mathbf{k}_\perp} \phi_n(\mathbf{k}_\perp)$$

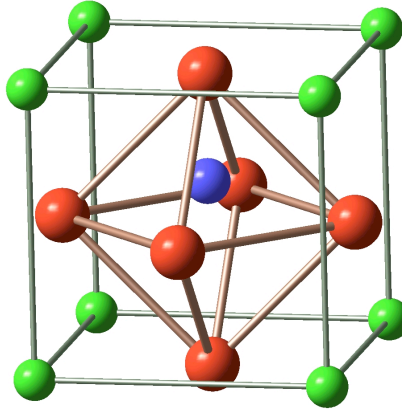


# Sample Application: Born $Z^*$

$$Z_{j\alpha\beta}^* = \frac{dP_\alpha}{dR_{j\beta}} \simeq \frac{\Delta P_\alpha}{\Delta R_{j\beta}}$$



*Paraelectric*



*Ferroelectric*

$$Z^*(\text{Ba}) = +2 e \quad ?$$

$$Z^*(\text{Ti}) = +4 e \quad ?$$

$$Z^*(\text{O}_\text{I}) = -2 e \quad ?$$

$$Z^*(\text{O}_\text{II}) = -2 e \quad ?$$

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# Wannier function representation

Choose Wannier functions as

$$w_n(\mathbf{r} - \mathbf{R}) = \int_{BZ} \Psi_{n\mathbf{k}}(\mathbf{r}) e^{-i\mathbf{k} \cdot \mathbf{R}} d\mathbf{k}$$

or, more precisely,

$$w_n(\mathbf{r} - \mathbf{R}) = \int_{BZ} e^{i\varphi_n(\mathbf{k})} \Psi_{n\mathbf{k}}(\mathbf{r}) e^{-i\mathbf{k} \cdot \mathbf{R}} d\mathbf{k}$$

or, more precisely,

$$w_n(\mathbf{r} - \mathbf{R}) = \int_{BZ} \sum_m U_{mn}^{(\mathbf{k})} \Psi_{m\mathbf{k}}(\mathbf{r}) e^{-i\mathbf{k} \cdot \mathbf{R}} d\mathbf{k}$$

in order to minimize

$$\Omega = \sum_n [\langle r^2 \rangle_n - \bar{\mathbf{r}}_n^2]$$

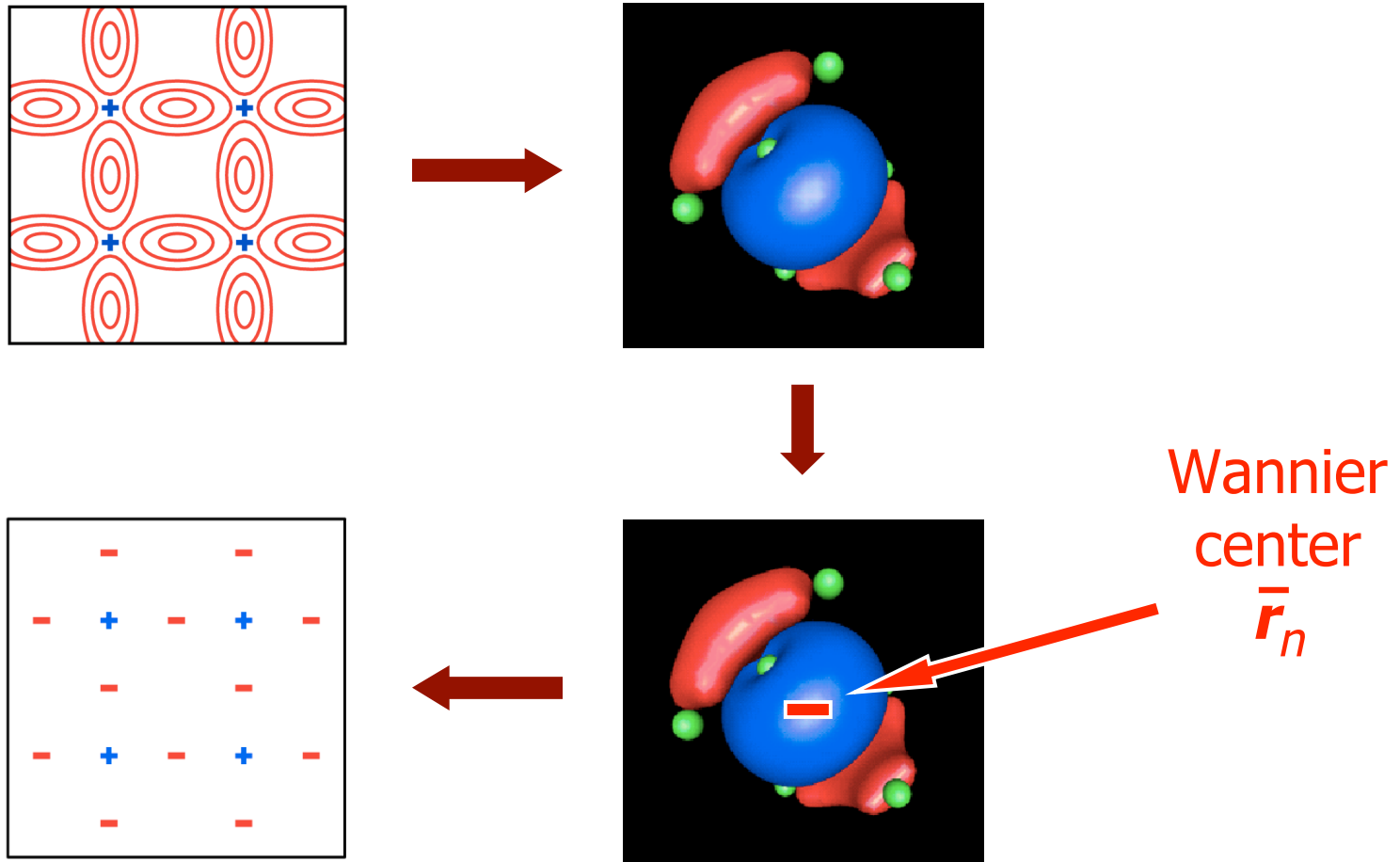
*(Marzari and Vanderbilt,  
1997)*

Here  $\langle r^2 \rangle_n = \langle 0n | r^2 | 0n \rangle$ ,

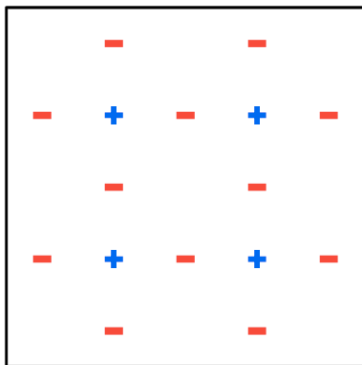
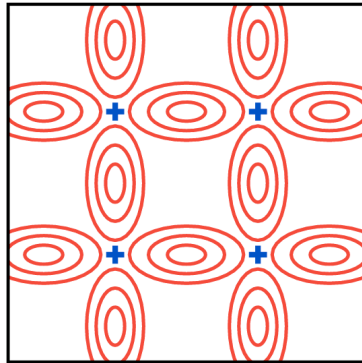
$$\bar{\mathbf{r}}_n = \langle 0n | \mathbf{r} | 0n \rangle$$

“Wannier center”

# Mapping to Wannier centers



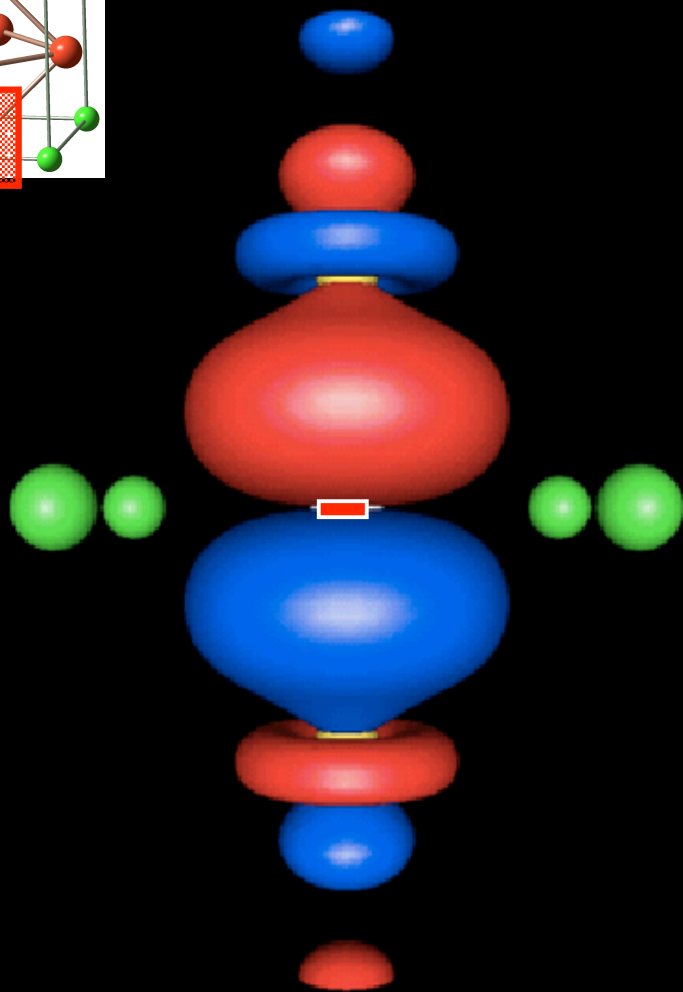
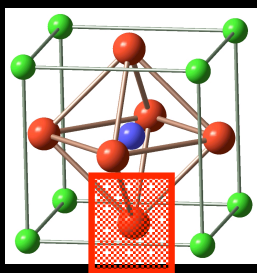
# Mapping to Wannier centers



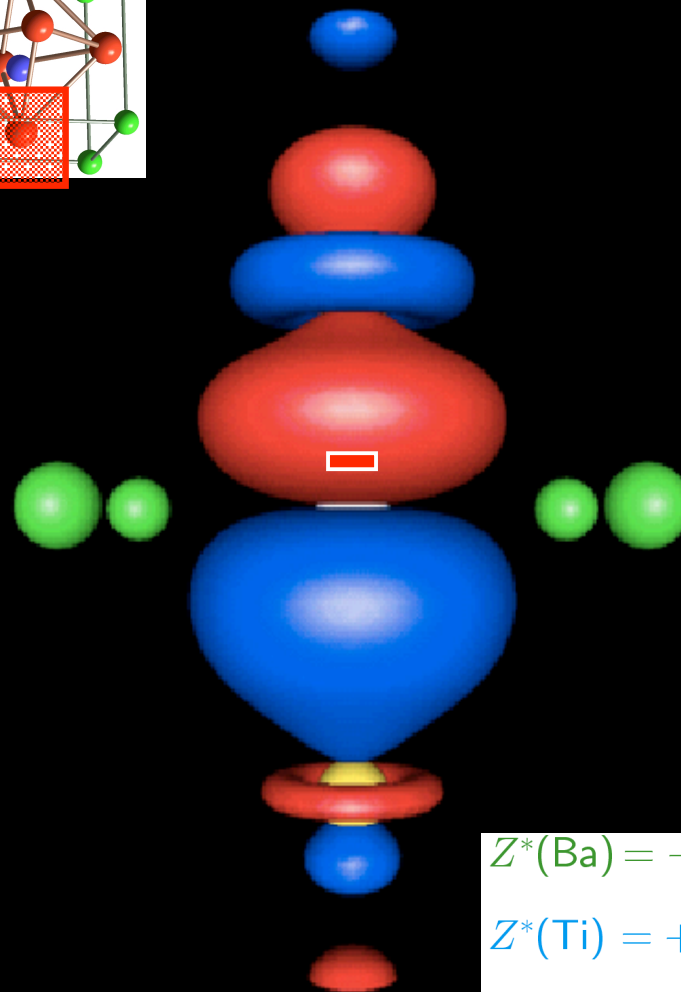
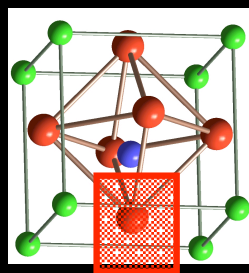
## Wannier dipole theorem

$$\langle \mathbf{P} \rangle = \sum_{ion} (Z_{ion} e) \langle \mathbf{r}_{ion} \rangle + \sum_{wf} (-2e) \langle \mathbf{r}_{wf} \rangle$$

- Exact!
- Gives local description of dielectric response!



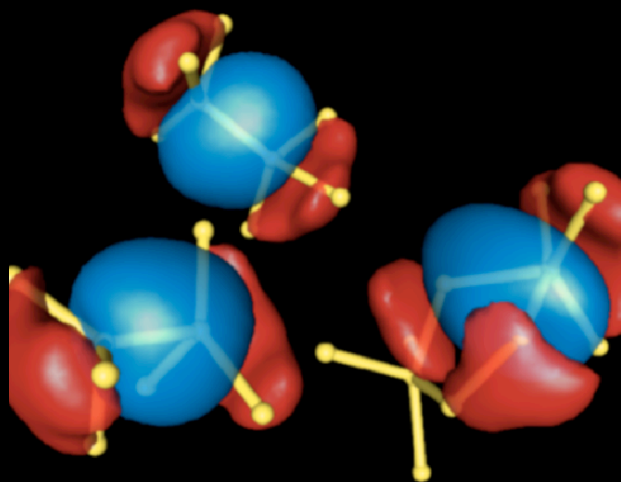
Ferroelectric BaTiO<sub>3</sub>



$$\begin{aligned} Z^*(\text{Ba}) &= +2.75 \, e \\ Z^*(\text{Ti}) &= +7.16 \, e \\ Z^*(\text{O}_\text{I}) &= -5.69 \, e \\ Z^*(\text{O}_\text{II}) &= -2.11 \, e \end{aligned}$$

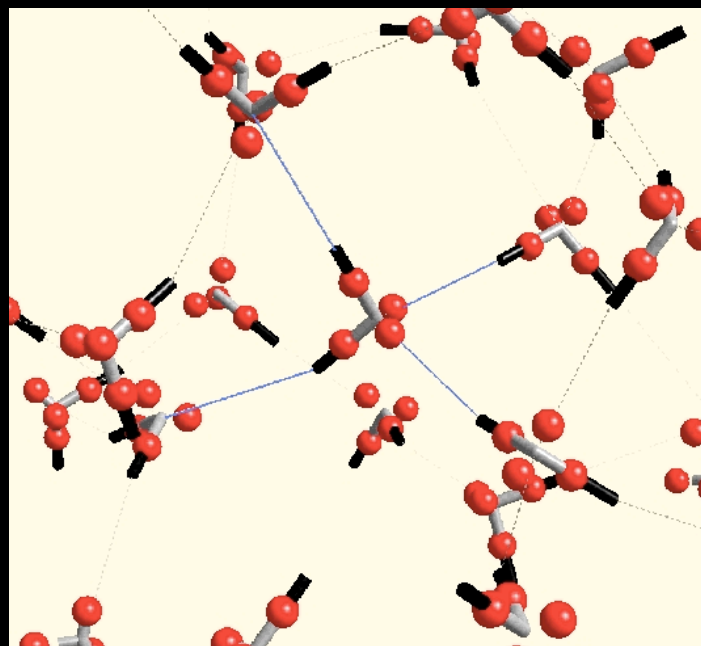
(Courtesy N. Marzari)

## Wannier functions in *a*-Si



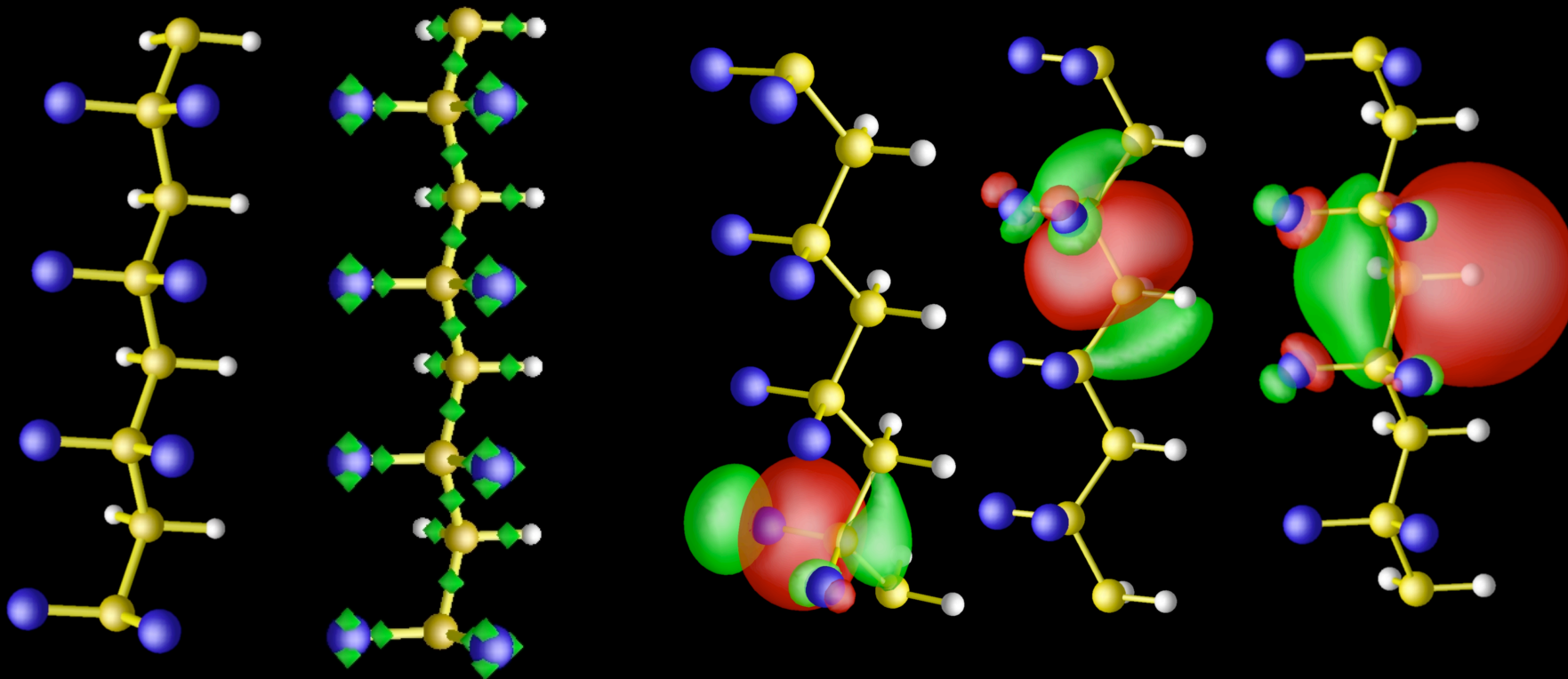
*Fornari et al.*

## Wannier functions in *l*-H<sub>2</sub>O



*Silvestrelli et al.*

# Wannier analysis of PVDF polymers and copolymers



Courtesy S. Nakhmanson



## Note upcoming release of public max-loc Wannier code...

 <http://www.wannier.org/>

**WANNIER90**

[Home](#) [History](#) [Download](#) [User Guide](#) [Forum](#) [Theory](#) [Gallery](#) [People](#)

**Welcome!**

This is the home of maximally-localised Wannier functions (MLWFs) and Wannier90, the computer program that calculates them. Wannier90 is released under the [GNU General Public License](#).

**Latest News**

**6 MAR 2006**

Expected release of Wannier90: April 2006

**12 FEB 2006**

Website has gone online

[Valid XHTML](#) | [Valid CSS](#) | Copyright © 2006 Wannier Developers' Group

(Organized by Nicola Marzari)

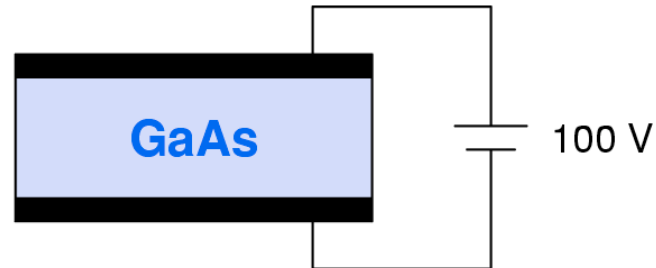
# Outline of Talk

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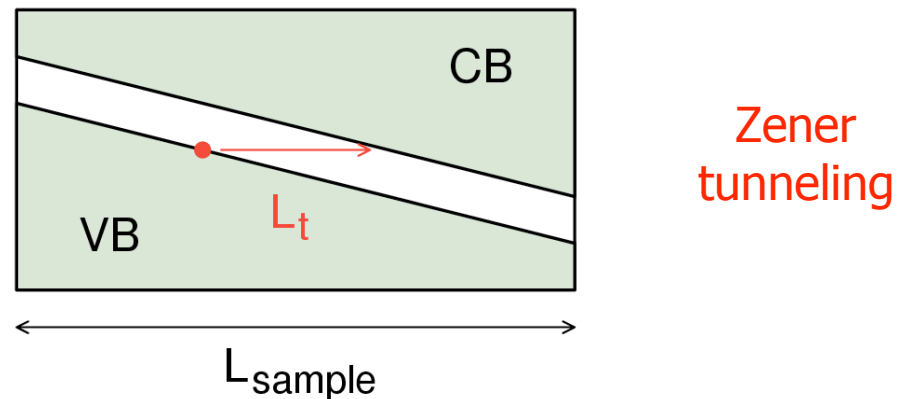
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# Electric Fields: The Problem

Easy to do in practice:



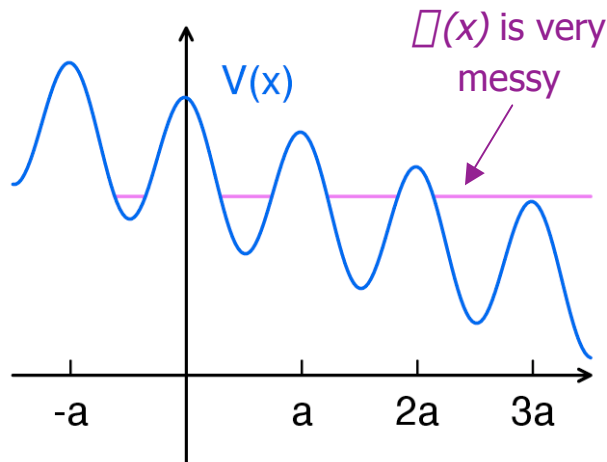
But ill-defined in principle:



For small E-fields,  $\lambda_{\text{Zener}} \gg \lambda_{\text{Universe}}$  ; is it OK?

# Electric Fields: The Problem

$$\begin{aligned} H &= H_0 - e \mathcal{E} x \\ &= \frac{P^2}{2m} + \tilde{V}(x), \quad \tilde{V}(x) = V_{\text{per}}(x) - e \mathcal{E} x \end{aligned}$$



- $\tilde{V}(x)$  is not periodic
- Bloch's theorem does not apply

# Electric Fields: The Solution

- Seek **long-lived resonance**
- Described by **Bloch functions**
- Minimizing the **electric enthalpy** functional

$$F = E - \mathcal{E} \cdot \mathbf{P} \quad (\text{Nunes and Gonze, 2001})$$

$$E = \sum_{nk} \langle \psi_{nk} | T + V_{\text{per}} | \psi_{nk} \rangle \quad \leftarrow \text{Usual } E_{\text{KS}}$$

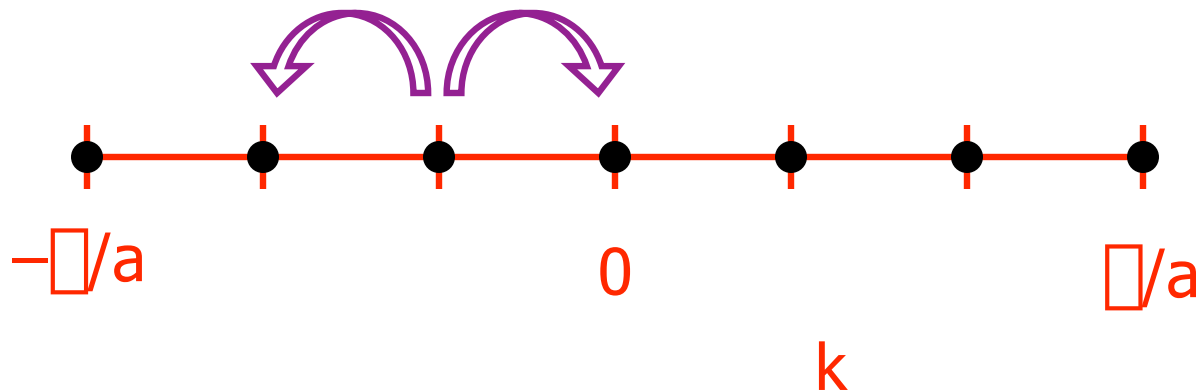
$$\mathbf{P} = \mathbf{P}[\hat{n}] = \mathbf{P}[\{ \psi_{nk} \}] \quad \leftarrow \text{Berry phase polarization}$$

Souza, Iniguez, and Vanderbilt, PRL 89, 117602 (2002);  
P. Umari and A. Pasquarello, PRL 89, 157602 (2002).

# Electric Fields: Implementation

As long as  $\Delta k$  is not too small:

- Can use standard methods to find minimum
- The  $\Delta$  P term introduces coupling between k-points

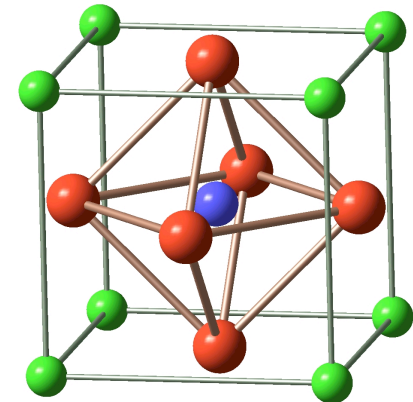


# Sample Application: Born $Z^*$

$$Z_{j\alpha\beta}^* = \frac{dP_\alpha}{dR_{j\beta}} \simeq \frac{\Delta P_\alpha}{\Delta R_{j\beta}}$$

$$Z_{j\alpha\beta}^* = \frac{dF_{j\beta}}{d\mathcal{E}_\alpha} \simeq \frac{\Delta F_{j\beta}}{\Delta \mathcal{E}_\alpha}$$

$$\text{(Relation: } Z_{j\alpha\beta}^* = - \frac{d^2 E}{d\mathcal{E}_\alpha dR_{j\beta}} \text{)}$$



Can check that previous results  
for BaTiO<sub>3</sub> are reproduced

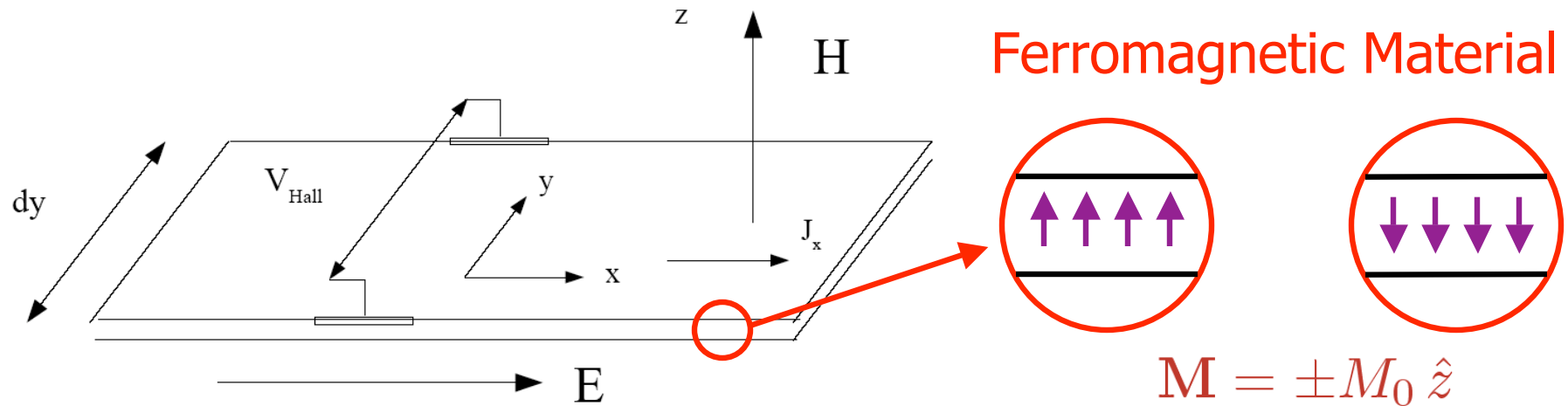
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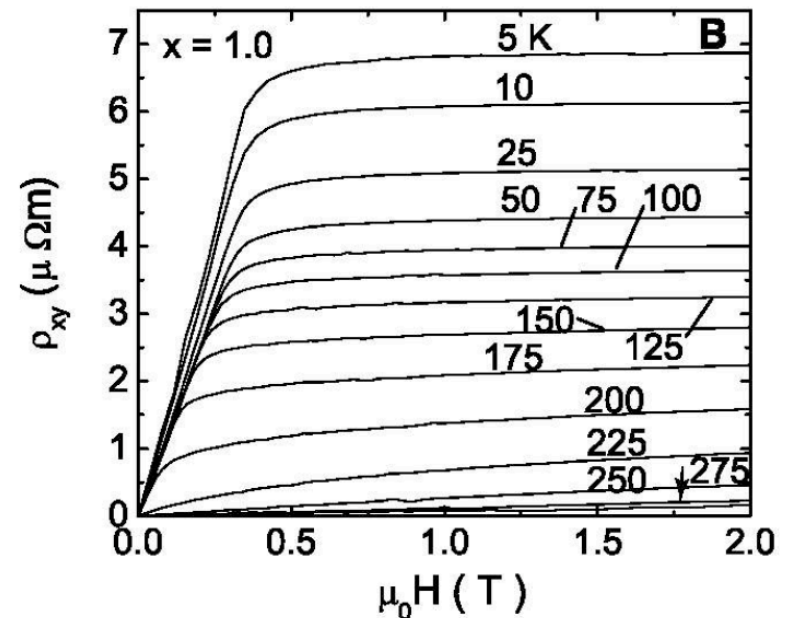
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# Anomalous Hall effect



$$\sigma_{xy}^{\text{Hall}} = R_0 B_z \pm \sigma_{xy}^{\text{AHE}}$$



$\text{CuCr}_2\text{Se}_{4-x}\text{Br}_x$  Wei-Li Lee et al. Science (2004)

# Anomalous Hall effect

- Karplus-Luttinger theory (1954)
  - Scattering-free, intrinsic
- Skew-scattering mechanism (1955)
  - Impurity scattering
- Side-jump mechanism (1970)
  - Impurity or phonon scattering
- Berry-phase theory (1999)
  - Restatement of Karplus-Luttinger

Semiclassical equations  
of motion:

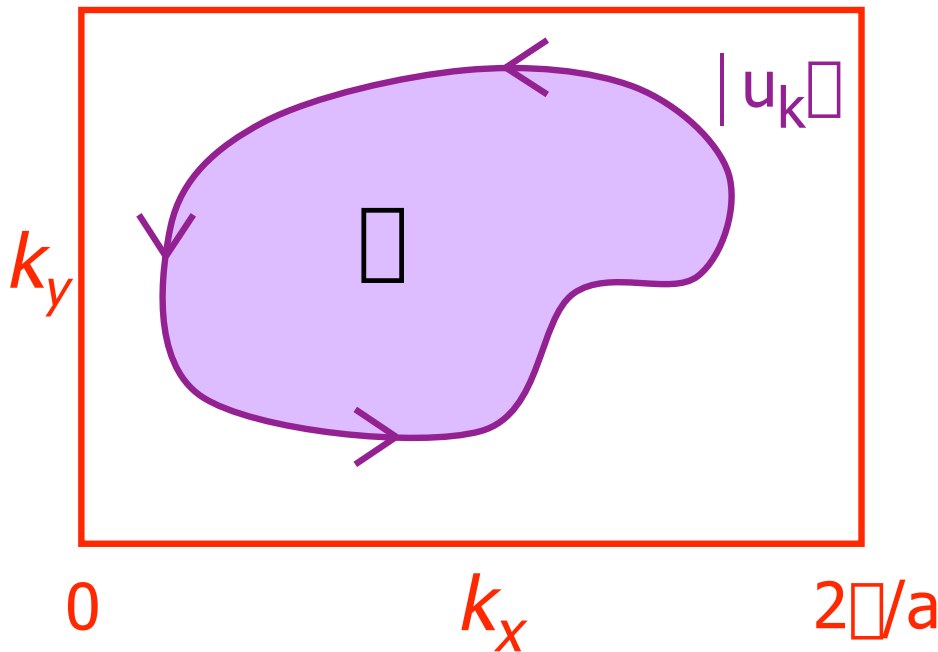
$$\dot{\mathbf{r}} = \frac{1}{\hbar} \nabla_{\mathbf{k}} \mathcal{E}_n - \dot{\mathbf{k}} \times \Omega_n(\mathbf{k})$$

$$\hbar \dot{\mathbf{k}} = -e\mathbf{E} - e\dot{\mathbf{r}} \times \mathbf{B}$$

Sundaram and Niu, PRB 59,  
14925 (1999).

$$\sigma_{xy}^{\text{AHE}} = \frac{-e^2 d}{(2\pi)^3 \hbar} \int d^3 k \Omega_z(\mathbf{k})$$

# Stokes theorem: Berry curvature



Berry potential:

$$\mathbf{A}(\mathbf{k}) = -\text{Im} \langle u_{\mathbf{k}} | \nabla_{\mathbf{k}} | u_{\mathbf{k}} \rangle$$

Berry phase:

$$\phi = \oint \mathbf{A}(\mathbf{k}) \cdot d\mathbf{k}$$

Berry curvature:

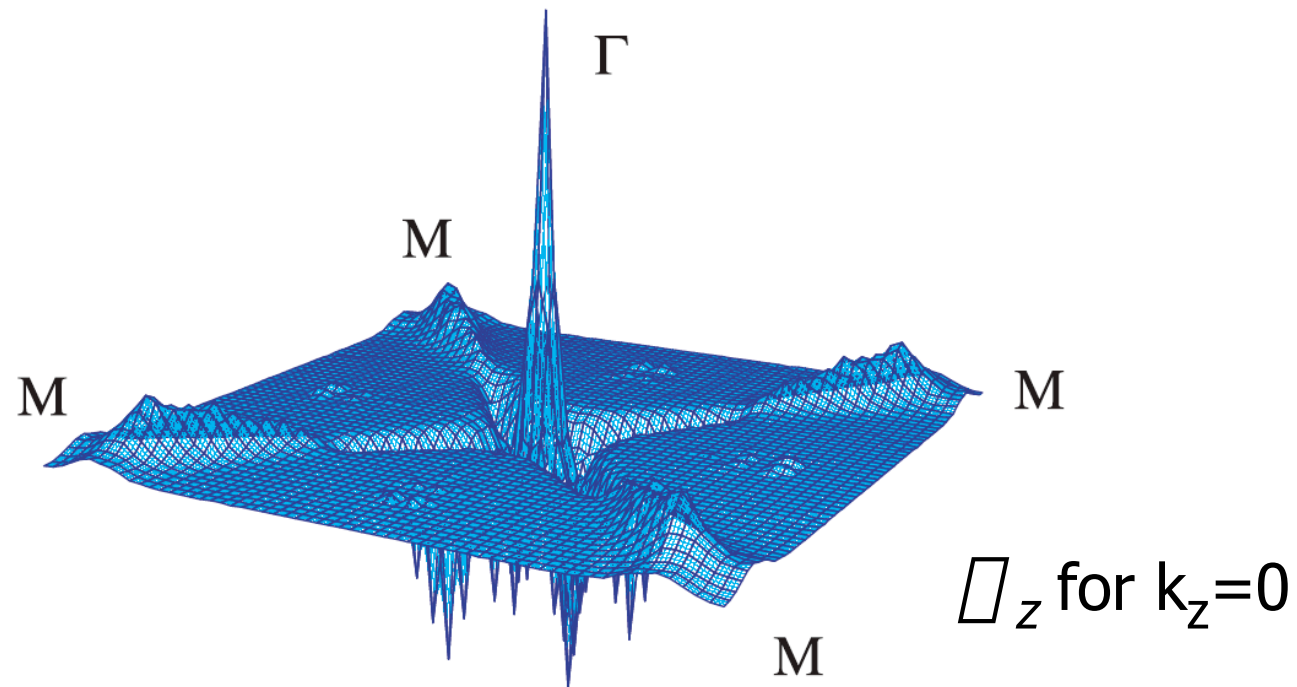
$$\Omega(\mathbf{k}) = \nabla \times \mathbf{A}$$

Stoke's theorem:

$$\phi = \int \Omega_z(\mathbf{k}) d^2k$$

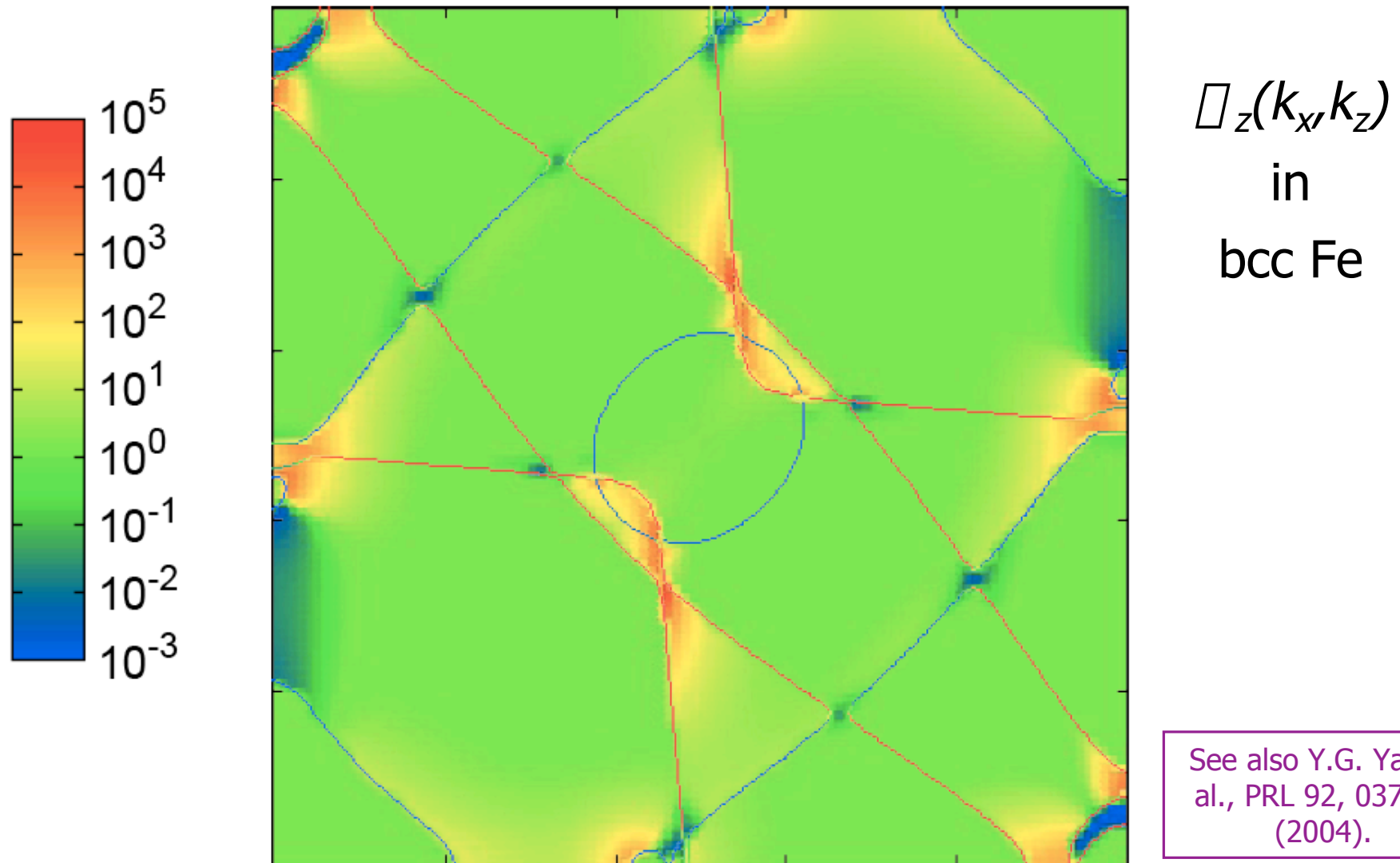
$$\Omega_z(\mathbf{k}) = -2\text{Im} \left\langle \frac{du}{dk_x} \left| \frac{du}{dk_y} \right. \right\rangle$$

## Anomalous Hall conductivity of $\text{SrRuO}_3$



Z. Fang et al, Science 302,  
92 (2003).

X. Wang, J. Yates, I. Souza, and D. Vanderbilt, G23.00001  
(Tuesday 8am).

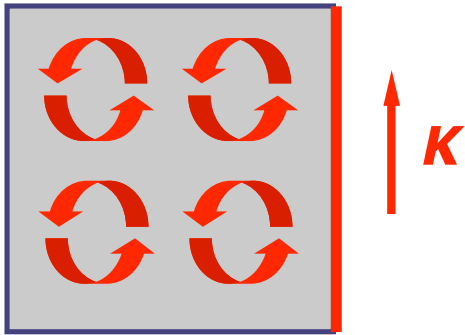


# Outline of Talk

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- Introduction
- Berry phases, potentials, and curvatures
- Realizations:
  - Electric polarization
  - Wannier functions
  - Electric fields
  - Anomalous Hall conductivity
  - **Orbital magnetization**
- Summary and prospects

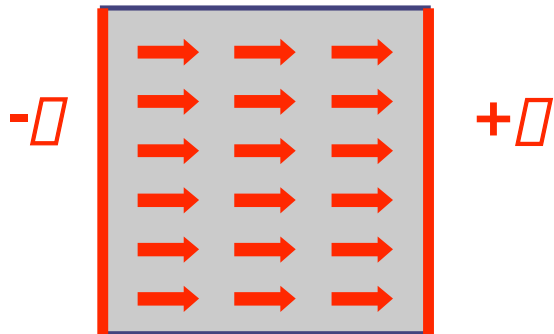
# Orbital Magnetization



**$M$**  is a bulk property?

$$\square \quad \mathbf{K} = \mathbf{M} \times \mathbf{n}$$

**$K$**  is *only apparently* a surface property?



**$P$**  is a bulk property

$$\square \quad \square = \mathbf{P} \cdot \mathbf{n}$$

$\square$  is *only apparently* a surface property

# Theory of orbital magnetization

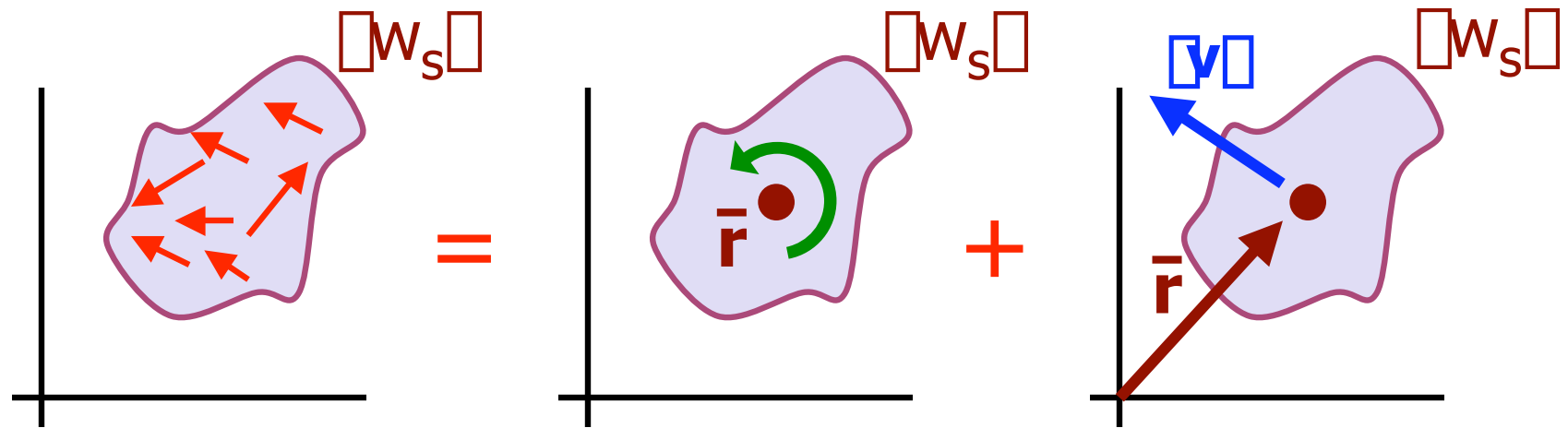
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T. Thonhauser, D. Ceresoli, D. Vanderbilt, and R. Resta,  
Phys. Rev. Lett. 95, 137205 (2005).

- Context:
  - Ferromagnetic insulators
  - Single-particle approximation
  - Vanishing magnetic field
- Used Wannier representation to derive a formula for the orbital magnetization



# Orbital currents in Wannier representation



$$\langle w_s | \mathbf{r} \times \mathbf{v} | w_s \rangle = \underbrace{\langle w_s | (\mathbf{r} - \bar{\mathbf{r}}) \times \mathbf{v} | w_s \rangle}_{\text{Local Circulation (LC)}} + \underbrace{\bar{\mathbf{r}} \times \langle w_s | \mathbf{v} | w_s \rangle}_{\text{Itinerant Circulation (IC)}}$$

Local Circulation  
(LC)

Itinerant Circulation  
(IC)

T. Thonhauser, H6.00001 (Tuesday 11:15am)  
(invited talk)

$$M = M_{\text{LC}} + M_{\text{IC}}$$

$$M_{\text{LC}} = \frac{-q}{\hbar c} \text{Im} \int_{\text{BZ}} \frac{d^2 k}{(2\pi)^2} \left\langle \frac{\partial u_{\mathbf{k}}}{\partial k_x} \middle| H_{\mathbf{k}} \middle| \frac{\partial u_{\mathbf{k}}}{\partial k_y} \right\rangle$$

Something new

$$M_{\text{IC}} = \frac{q}{2\hbar c} \int \frac{d^2 k}{(2\pi)^2} E(\mathbf{k}) \Omega(\mathbf{k})$$

Berry curvature

See also D. Xiao, J. Shi and Q. Niu, PRL 95, 137204 (2005).

# Summary and Prospects

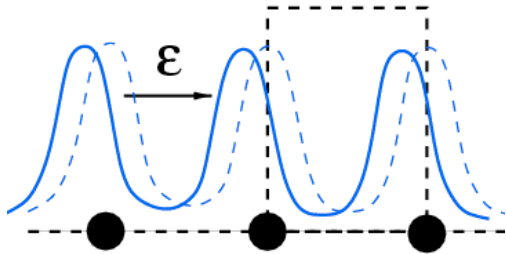
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- Berry phases are everywhere!
- We discussed:
  - Electric polarization
  - Electric fields
  - Anomalous Hall coefficient
  - Orbital magnetization
- Other “hot topics”:
  - Multiferroics and magnetoelectric effects
  - Single graphene sheets
  - Spin Hall effect and spin injection
- More Berry phases lurking around the corner?

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# Extras

# Electric Fields: Justification



Seek  
long-lived  
metastable  
periodic  
solution

- Want periodic charge density:

$$\rho(\mathbf{r}) = \rho(\mathbf{r} + \mathbf{R})$$

- Want periodic one-particle density matrix:

$$n(\mathbf{r}, \mathbf{r}') = n(\mathbf{r} + \mathbf{R}, \mathbf{r}' + \mathbf{R})$$

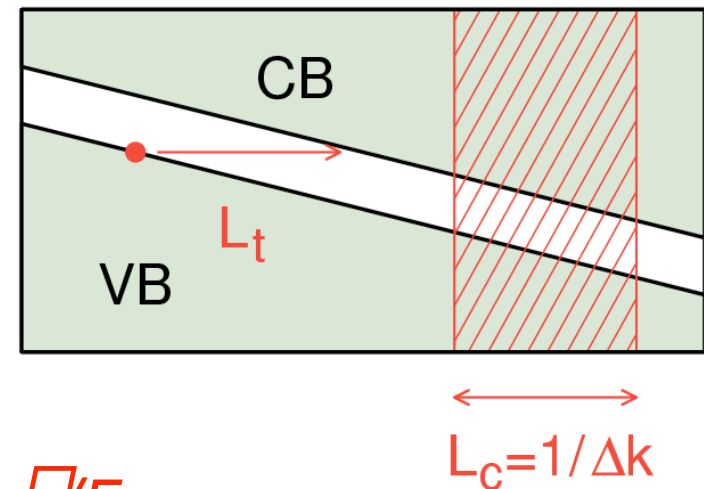
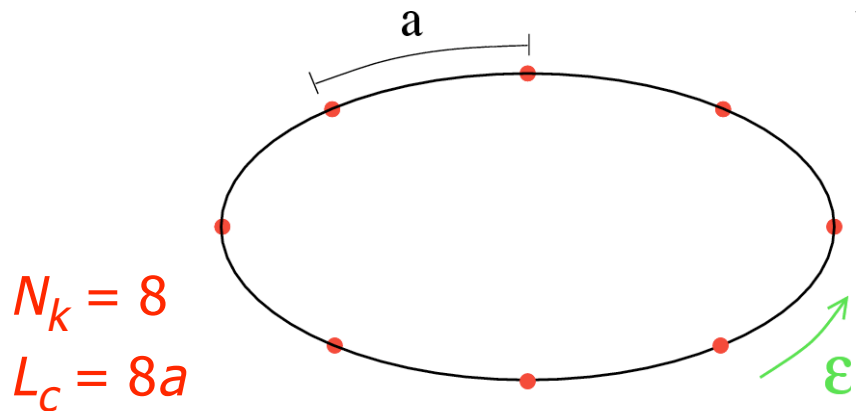
- Use Bloch representation of density matrix:

$$n(\mathbf{r}, \mathbf{r}') = \sum_{n\mathbf{k}} \psi_{n\mathbf{k}}^*(\mathbf{r}) \psi_{n\mathbf{k}}(\mathbf{r}')$$

even though  $\psi_{n\mathbf{k}}$  are not eigenstates!

# Electric Fields: The Hitch

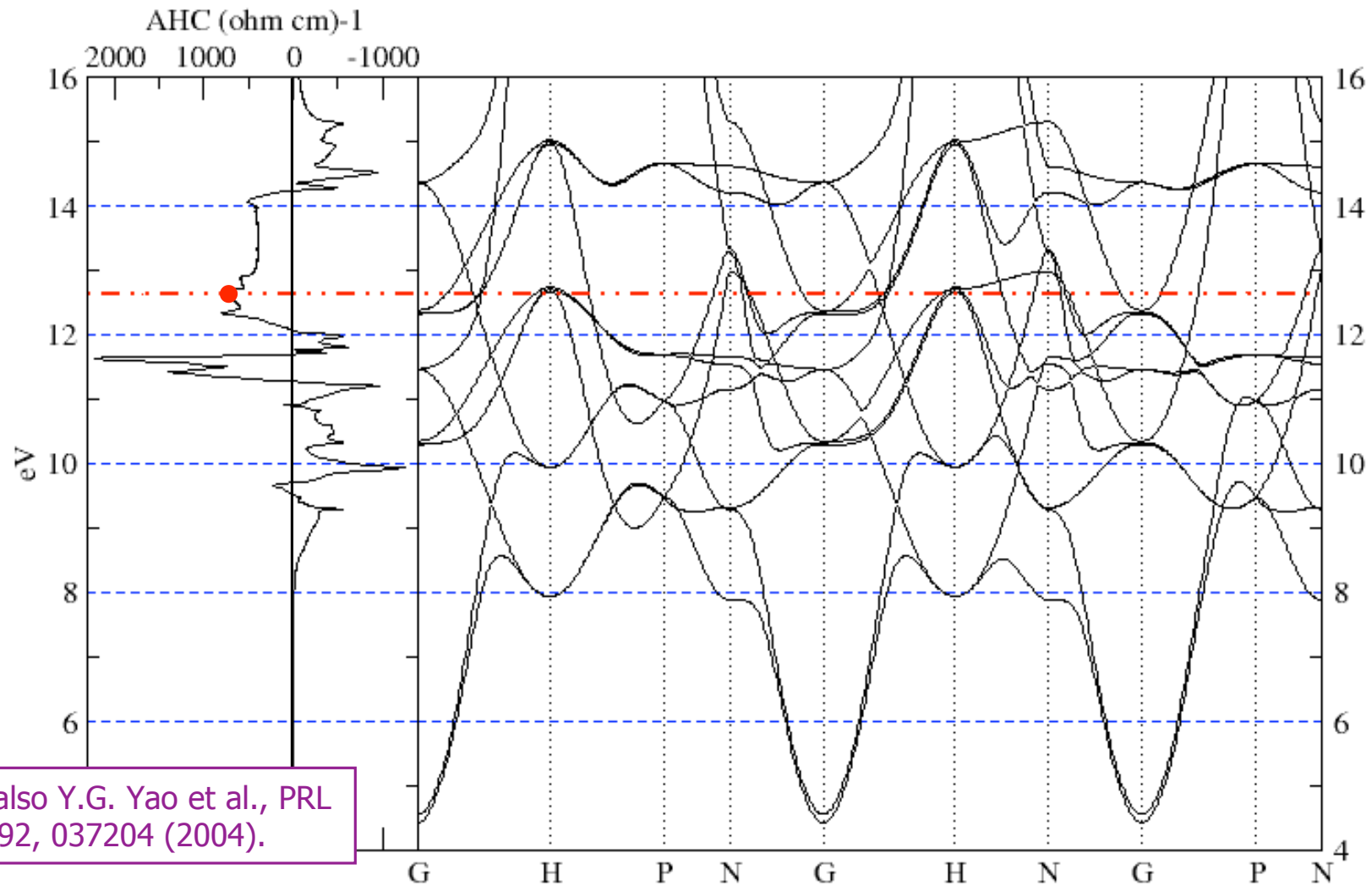
- There is a hitch!
- For given E-field, there is a limit on k-point sampling
- Length scale  $L_C = 1/\Delta k$
- Meaning:  $L_C$  = supercell dimension



- Solution: Keep  $\Delta k > 1/L_t = \epsilon/E_g$

X. Wang, J. Yates, I. Souza, and D. Vanderbilt, G23.00001  
(Tuesday 8am).

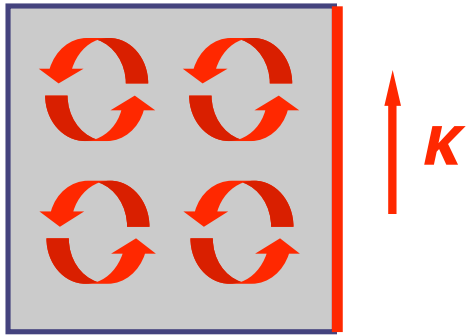
## Anomalous Hall conductivity of bcc Fe



See also Y.G. Yao et al., PRL  
92, 037204 (2004).

# Orbital Magnetization

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$$\mathbf{K} = \mathbf{M} \times \mathbf{n}$$

Is  $\mathbf{M}$  a *bulk property*?

Is  $\mathbf{K}$  *only apparently* a surface property?

## Definition:

If  $\mathbf{K}$  is predetermined at all surfaces in such a way that  $\mathbf{K} = \mathbf{M} \times \mathbf{n}$  for some vector  $\mathbf{M}$ , then  $\mathbf{M}$  is the bulk magnetization.



# Orbital Magnetization

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## Clarification:

- Microscopic  $\mathbf{M}(\mathbf{r})$  defined via  $\nabla \times \mathbf{M}(\mathbf{r}) = \mathbf{J}(\mathbf{r})$
- $\mathbf{M}(\mathbf{r})$  ill-defined:  $\mathbf{M}(\mathbf{r}) \nabla \cdot \mathbf{M}(\mathbf{r}) + \mathbf{M}_0 + \nabla \nabla$
- Therefore, cannot define  $\mathbf{M}$  as cell average of  $\mathbf{M}(\mathbf{r})$

Conclusion:  $\mathbf{M}$  is not, even in principle, a functional of the bulk current distribution  $\mathbf{J}(\mathbf{r})$

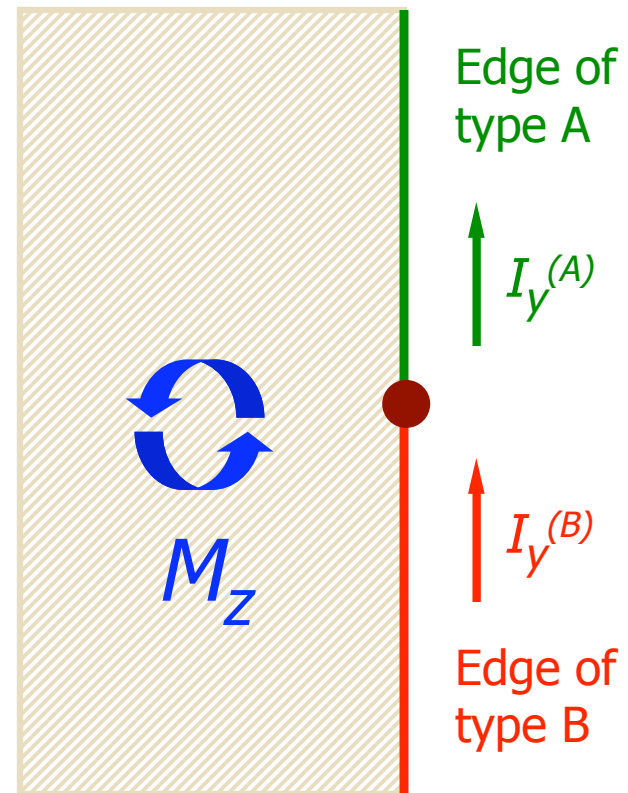
*(Hirst, RMP, 1997)*

Just as:  $\mathbf{P}$  is not, even in principle, a functional of the bulk charge density distribution  $\rho(\mathbf{r})$

# Strong reasons to expect bulk $M$

- Nearsightedness:  
Surface current depends only  
on local environment
- Stationary quantum state:  
 $d\rangle/dt = 0$
- Conservation of charge:  
 $\nabla \cdot \mathbf{J} = 0$

So:  $I_y^{(A)} = I_y^{(B)} = M_z$



# Comparison: ***P*** vs. ***M***

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## Electric Polarization

Defined for insulators only

$\mathbf{P}(\mathbf{r})$  insufficient in principle;  
need access to Berry physics

$\mathbf{r}$  operator

“Quantum of polarization”

Derivable from adiabatic theory

Derivable from Wannier rep.

## Orbital Magnetization

Insulators and metals  
with broken TR symmetry

$\mathbf{J}(\mathbf{r})$  insufficient in principle;  
need access to Berry physics

$\mathbf{r} \times \mathbf{v}$  operator

No quantum (no monopoles)

No obvious adiabatic theory

Derivable from Wannier rep.?

# Ultrasoft Pseudopotentials

Then, the good news:

\*

| Table 4. The 10 Hot Papers, as Defined in the Text, Arranged Chronologically      |         |         |                                                                                              |                                                                            |
|-----------------------------------------------------------------------------------|---------|---------|----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| Publication                                                                       | # cites | Av. age | Title                                                                                        | Author(s)                                                                  |
| PR 40, 749 (1932)                                                                 | 562     | 55.8    | On the Quantum Correction for Thermodynamic Equilibrium                                      | E. Wigner                                                                  |
| PR 47, 777 (1935)                                                                 | 492     | 59.6    | Can Quantum-Mechanical Description of Physical Reality Be Considered Complete?               | A. Einstein, B. Podolsky, N. Rosen                                         |
| PR 109, 1492 (1958)                                                               | 871     | 32.0    | Absence of Diffusion in Certain Random Lattices                                              | P. W. Anderson                                                             |
| PR 136, B864 (1964)                                                               | 2460    | 28.7    | Inhomogeneous Electron Gas                                                                   | P. Hohenberg, W. Kohn                                                      |
| PR 140, A1133 (1965)                                                              | 3227    | 26.6    | Self-Consistent Equations Including Exchange and Correlation Effects                         | W. Kohn, L. J. Sham                                                        |
| PRB 13, 5188 (1976)                                                               | 1023    | 20.8    | Special Points for Brillouin-Zone Integrations                                               | H. J. Monkhorst, J. D. Pack                                                |
| PRL 48, 1425 (1982)                                                               | 829     | 15.1    | Efficacious Form for Model Pseudopotentials                                                  | L. Kleinman, D. M. Bylander                                                |
| PRB 41, 7892 (1990)                                                               | 691     | 9.7     | Soft Self-Consistent Pseudopotentials in a Generalized Eigenvalue Formalism                  | D. Vanderbilt                                                              |
| PRB 45, 13244 (1992)                                                              | 394     | 8.1     | Accurate and Simple Analytic Representation of the Electron-Gas Correlation Energy           | J. P. Perdew, Y. Wang                                                      |
| PRL 70, 1895 (1993)                                                               | 495     | 7.4     | Teleporting an Unknown Quantum State via Dual Classical and Einstein-Podolsky-Rosen Channels | C. H. Bennett, G. Brassard, C. Crépeau, R. Jozsa, A. Peres, W. K. Wootters |
| <i>PR, Physical Review; PRB, Physical Review B; PRL, Physical Review Letters.</i> |         |         |                                                                                              |                                                                            |

Sidney Redner, Physics Today, June 2005.

\* (A hot paper is...) "defined as a nonreview paper with 350 or more citations, an average ratio of citation age to publication age greater than two-thirds, and a citation rate increasing with time."

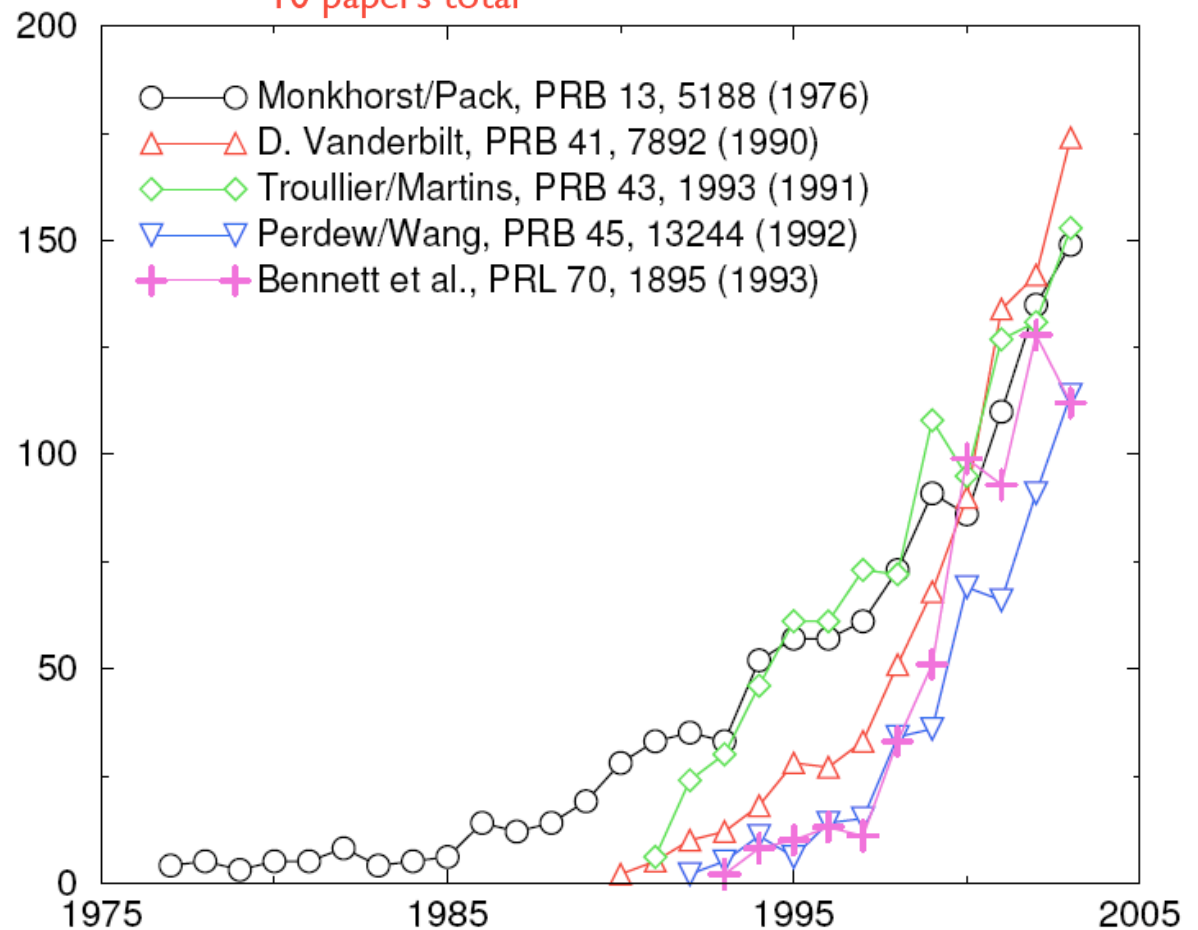
# Ultrasoft Pseudopotentials

Then, the good news:

Hot Publications

# cites > 350  
citation rate ↑

10 papers total



Sidney Redner, APS talk,  
March, 2004; Physics  
Today, June 2005.